

Varendra University
Department of Computer Science and Engineering

Project Report

Title: Fire and Smoke Alarm system with SMS Notification
Course Title: Microcontroller, Computer Peripherals and Interfacing Lab
Course Code: CSE 426

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Acknowledgement

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Abstract

This project presents a Fire and Smoke Alarm System with SMS Notification, designed to detect smoke, gas leakage, heat, and flame using multiple sensors. An Arduino Nano processes sensor data, displays system status on a 16×2 LCD, and triggers alerts through a GSM 800L module. The system immediately sends SMS notifications to the user when a hazardous condition is detected. A regulated 12V power supply ensures smooth operation using AN780509, buck converter, capacitors, and diodes. This project offers a low-cost, reliable early warning solution for home and industrial safety.

Introduction

Fire accidents can cause severe damage to life and property. Early detection is essential to minimize such losses. Traditional fire alarms only generate audible signals and often fail to alert the user when they are away. To address this limitation, the proposed system integrates fire detection with SMS notification capability.

Using sensors like MQ2 (smoke), gas sensor, temperature sensor, and fire sensor, the system continuously monitors the environment. When abnormal conditions occur, the system immediately alerts the user through SMS using the GSM 800L module. The design is costeffective, easy to implement, and suitable for both residential and industrial environments.

Literature Review

Several studies and projects have focused on fire detection systems using microcontrollers:

- **Early Fire Detection Systems (2016–2020):**
  Research shows that combining smoke and temperature sensors improves detection accuracy and reduces false alarms.
- **IoT-Based Fire Alarm Systems:**
  Recent works integrate WiFi modules (ESP8266) for online notification. However, these systems depend on internet availability.
- **GSM-Based Systems:**
  GSM modules like SIM800L offer reliable notifications without requiring internet access. Literature suggests they perform well in rural or remote areas.
- **Multisensor Approaches:**
  Research emphasizes that using multiple sensors smoke, fire, gas, and temperature provides more accurate hazard detection.

This project builds upon these studies by combining multiple sensors with GSM alerting to maximize reliability and user safety.

Methodology

The project follows these steps:

1. **Requirement analysis**: Identifying necessary sensors and components.
2. **Circuit design**: Creating a schematic including sensors, GSM and power supply.
3. **Programming**: Writing Arduino Code.
4. **Hardware assembly**: Connecting sensors, LCD, GSM and power supply.
5. **Testing & Calibration:**
    - Testing sensor responses for smoke, temperature, and flame.
    - Adjusting threshold levels for real-world accuracy.
6. **Evaluation** – Observing system performance and recording results.

System Design

Block Diagram

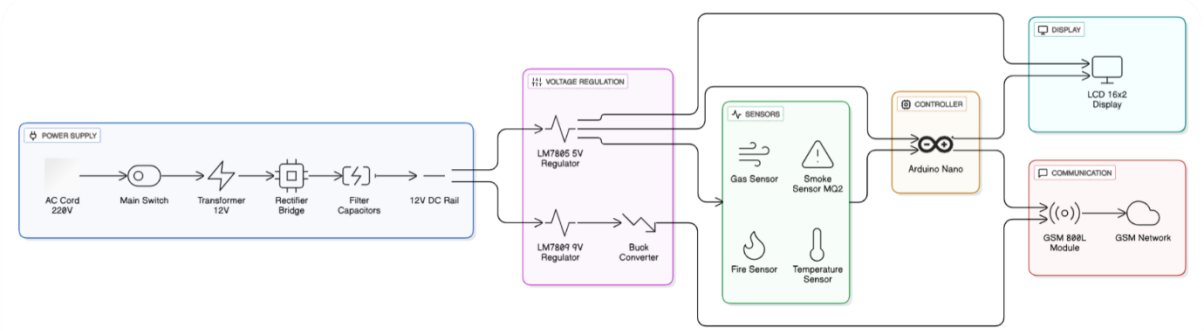


Figure: Block Diagram

Real-Time Images

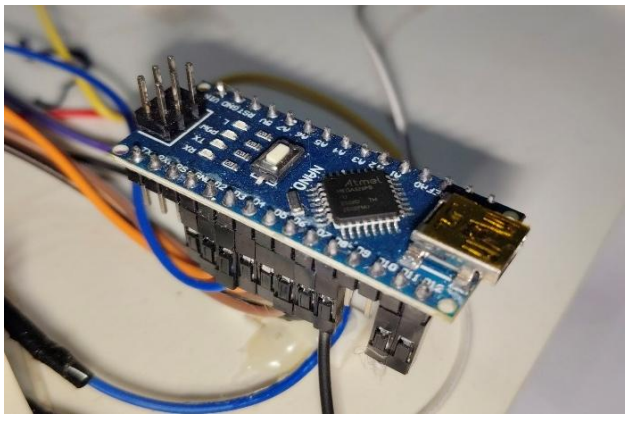


Figure: Arduino Nano

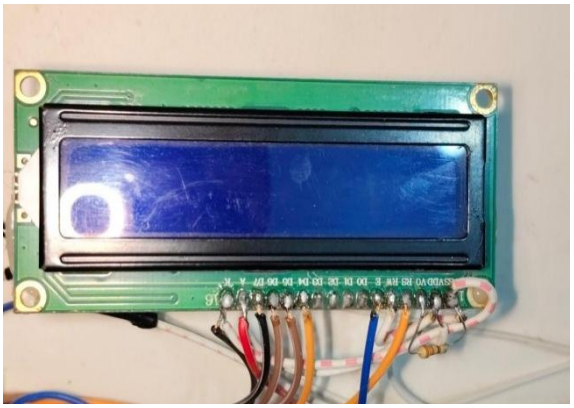


Figure: 16×2 LCD

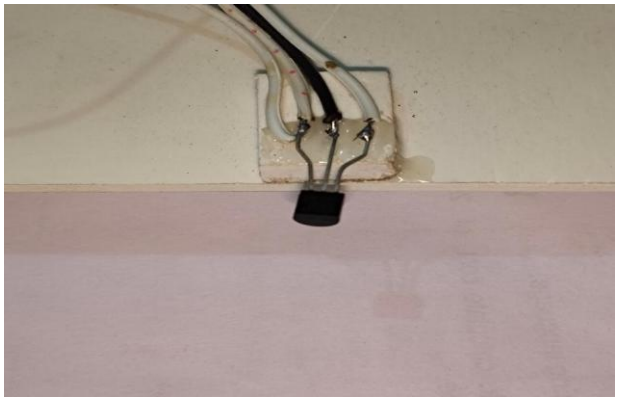


Figure: Temperature Sensor

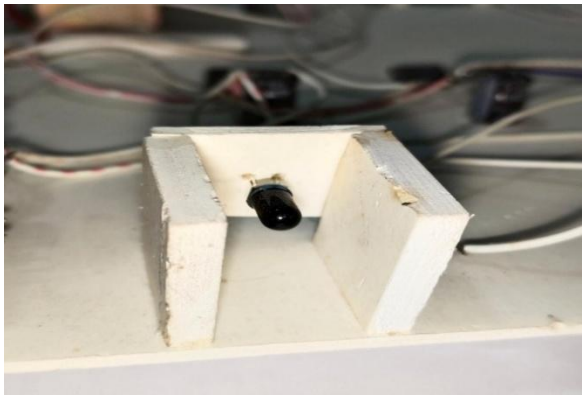


Figure: Fire Sensor



Figure: Resistor

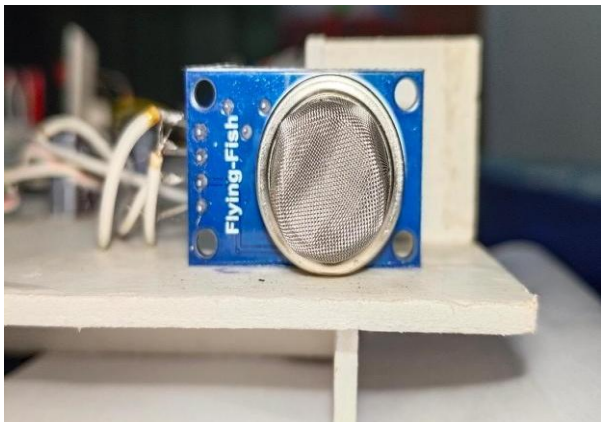


Figure: Smoke Sensor MQ2

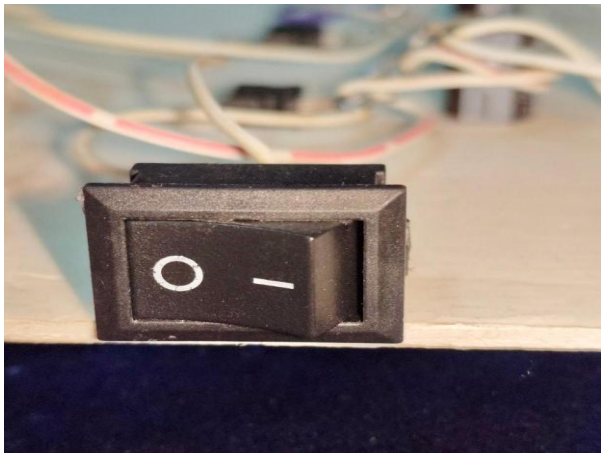


Figure: Switch

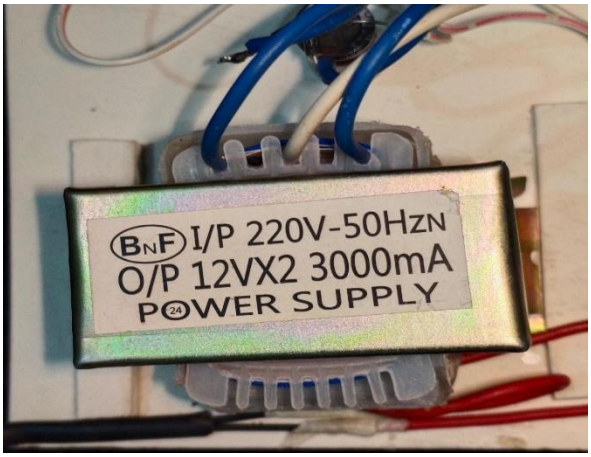


Figure: Transformer 12V

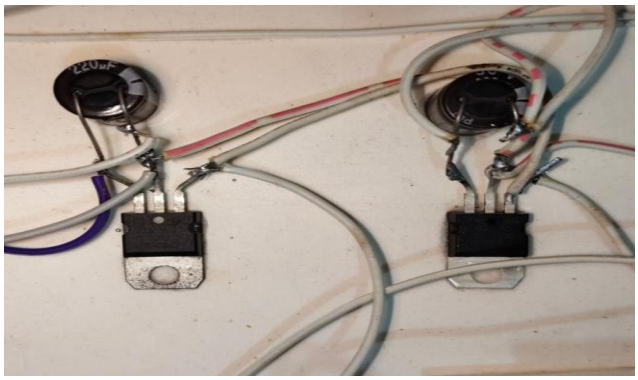


Figure: Regulator IC An7805-09



Figure: Capacitor

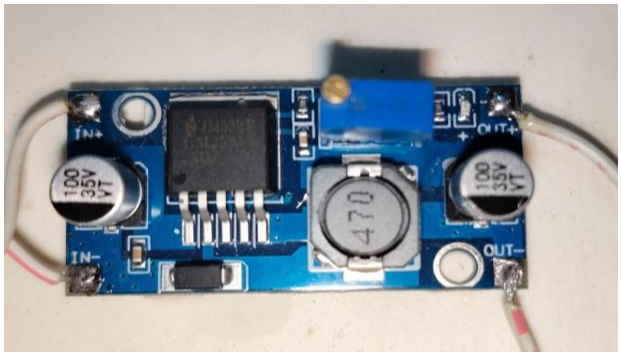


Figure: Buck Converter

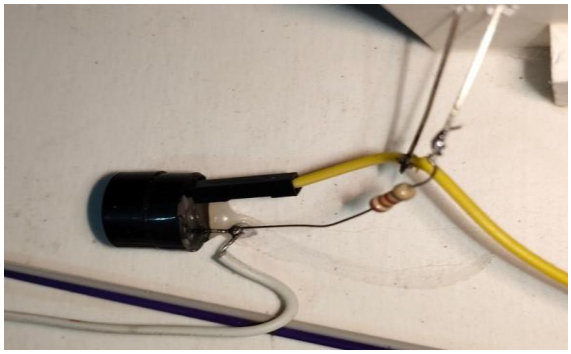


Figure: Buzzer

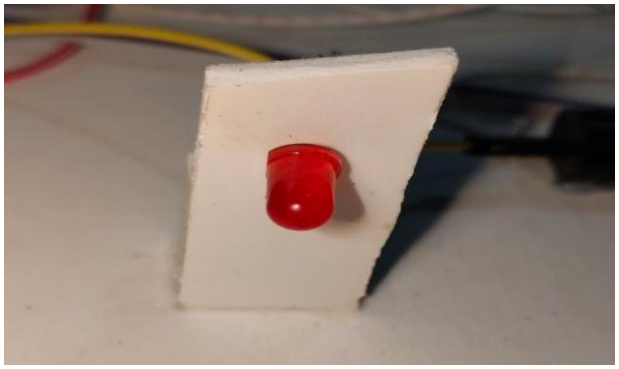


Figure: LED

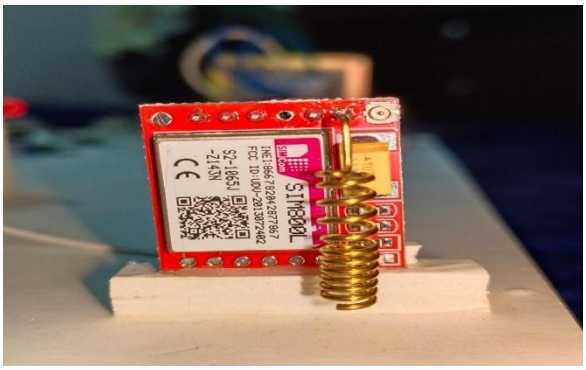


Figure: GSM 800L Module

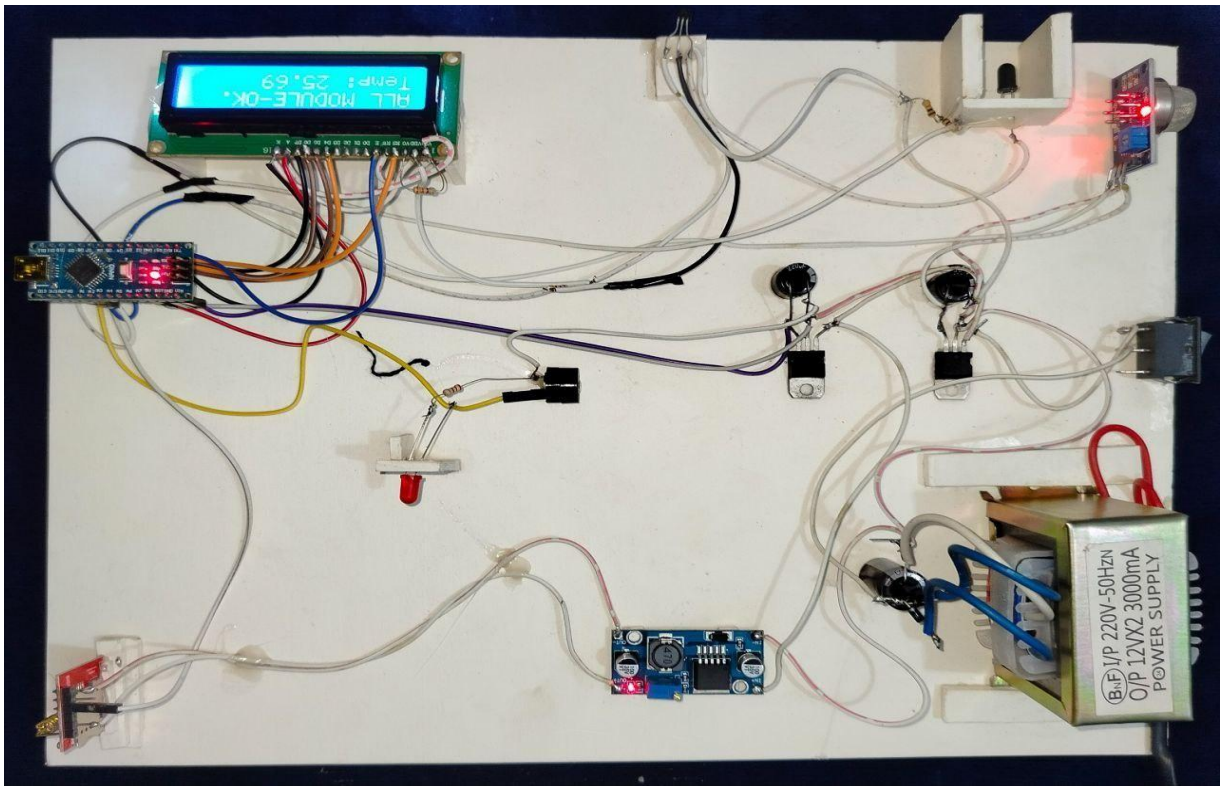


Figure: Full project in real-time

Cost Analysis

1	Transformer 12v	1pc	350/=
2	Arduino nano	1pc	480/=
3	Display LCD 16*2	1pc	170/=
4	Smoke Sensor MQ2	1pc	150/=
5	Fire Sensor	1pc	20/=
6	Temperature Sensor	1pc	140/=
7	Regulator IC An7805-09	2pc	30/=
8	Capacitor	3pc	30/=
9	Diode	2pc	5/=
10	Switch	1pc	10/=
11	Buck converter	1pc	120/=
12	Gsm 800L	1pc	550/=
13	Resistor	1pc	2/=
14	AC Cord	1pc	30/=
15	Buzzer	1pc	40/=
16	LED	1pc	5/=
		Total Tk	2132/=

Result Analysis

- The system successfully detected smoke, fire, gas leakage, and high temperature.
- LCD accurately displayed real-time sensor readings.
- The GSM 800L sent SMS alerts within 3–5 seconds after detection.
- No false alarms were observed during normal environmental conditions.
- Power supply remained stable with regulated 5V output.

Discussion

The system performed reliably under different conditions. Using multiple sensors enhanced accuracy, as smoke or temperature alone cannot confirm a fire. GSM-based alerting proved effective even without internet access. Sensor calibration was critical; improper thresholds could trigger false notifications. Future improvements may include IoT integration and mobile app monitoring.

Future Work

- Integrate WiFi/IoT (Blynk, MQTT) for mobile app monitoring.
- Add automatic water spray or extinguisher system.
- Include battery backup for uninterrupted operation.
- Add data logging to record sensor history.

Conclusion

The Fire and Smoke Alarm System with SMS Notification successfully detects hazardous conditions and immediately alerts the user with an SMS. This system enhances safety through early detection and is suitable for homes, offices, and industries. Its low cost and reliability make it a practical and effective fire-safety solution.

References

1. Arduino Documentation – https://www.arduino.cc
2. MQ2 Sensor Datasheet
3. SIM800L GSM Module Hardware Guide
4. Temperature and Fire Detection Research Papers (2018–2024)
5. Microcontroller-Based Safety Systems Journals

Work Progress Chart

Week	Tasks Completed
Week 1	Topic selection, component research, planning. Purchasing components, preparing circuit schematic.
Week 2	Writing basic Arduino code, testing sensors individually. Integrating sensors with LCD and Arduino.
Week 3	Adding GSM module and SMS alert functionality.
Week 4	Hardware assembly and power supply integration.
Week 5	Testing, debugging, threshold tuning. Final assembly, documentation, and report writing.

Improvements

- Adjusted smoke and gas sensor thresholds for stability
- Improved GSM signal reliability by adding a buck converter
- Used LCD to reduce wiring complexity
- Added diode protection to prevent reverse voltage damage